Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

Ronald Melvin R. Rosas — Marcus Rafael Tiongson

AI 201 — University of the Philippines, Diliman

Static Coefficients vs. Real-World Variance

Combine Harvester

$> 50\,\text{kW}$ · mobile · high field load

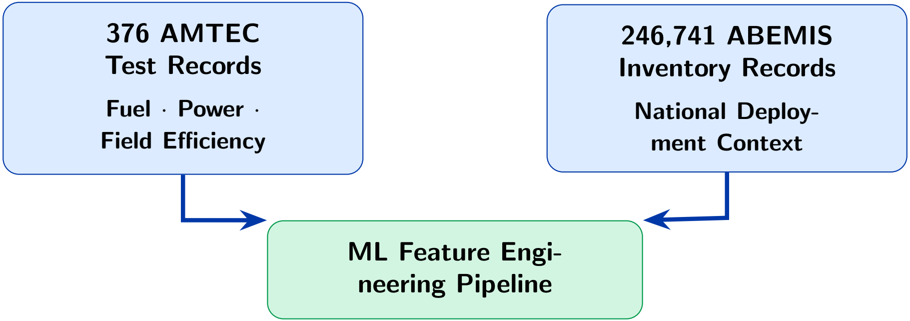
$\neq$

Stationary Water Pump

$< 5\,\text{kW}$ · fixed · low load

Deterministic formulas apply the same generalized constant to both.

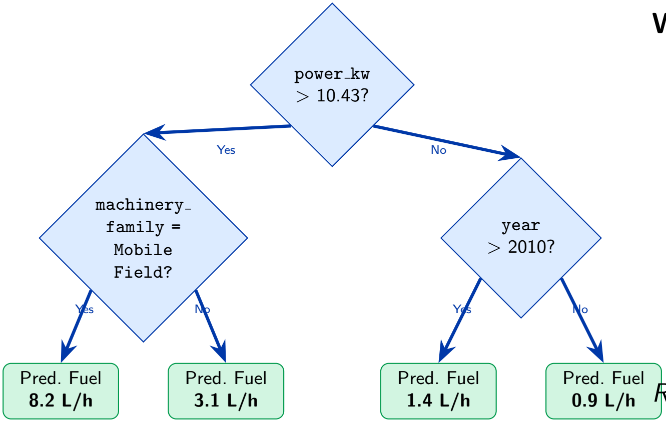
The Data: AMTEC & ABEMIS



AMTEC — standardized, controlled lab tests
Source: BAFE-AMTEC (1990s–2026)

ABEMIS — national deployment inventory
Coverage: all 17 Philippine regions

The Engine: Capturing Non-Linear Physical Interactions



Why not OLS?

- Power and fuel do *not* scale linearly across machine types
- Each tree partitions the feature space differently
- 300 trees average out variance — robust predictions
- No manual feature crossing required

$$R^2_{\text{RF}} = 0.916 \quad \text{vs.} \quad R^2_{\text{OLS}} = 0.704$$

Random Forest vs. OLS Baseline ($n_{\text{test}} = 76$)

Model	R^2	MAE (L/h)	RMSE (L/h)
OLS Global Linear	0.704	1.882	2.875
OLS Hierarchical Fallback	0.703	1.879	2.879
Random Forest	**0.916**	**0.950**	**1.534**

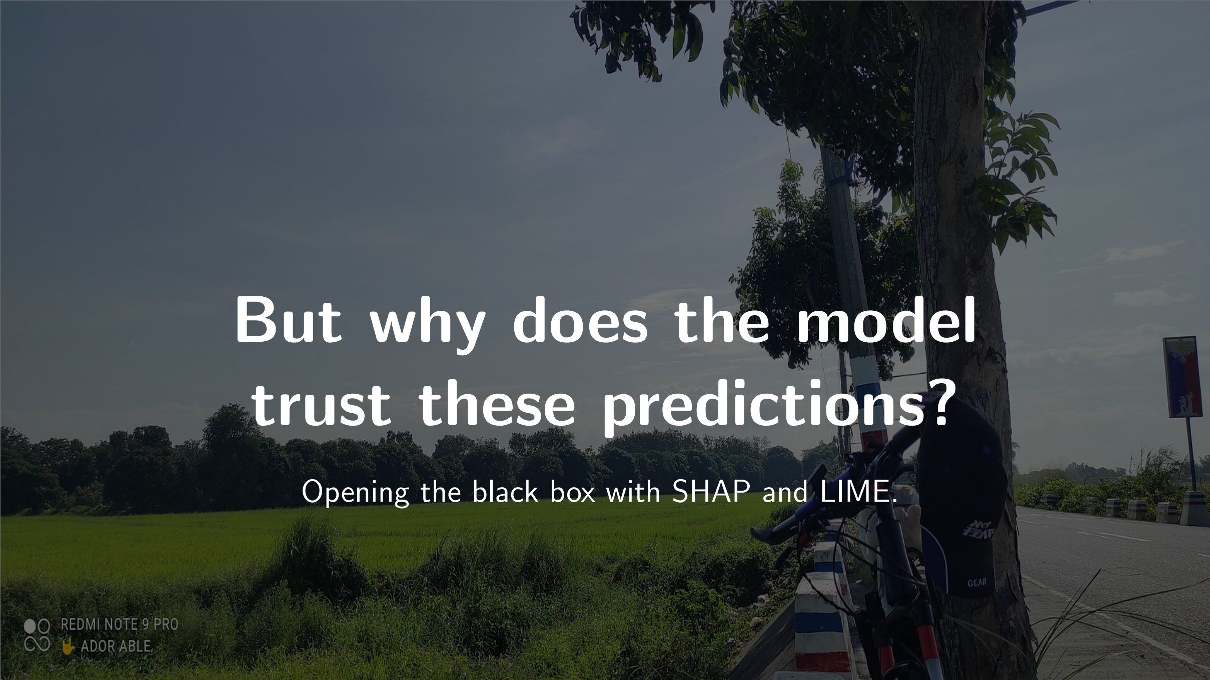
+30%
better R^2

2×
lower MAE

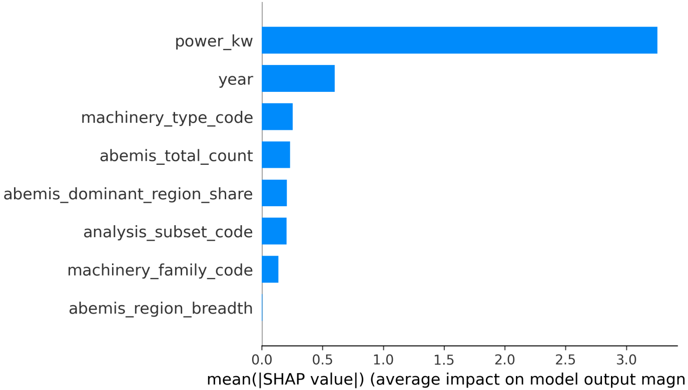
2×
lower RMSE

But why does the model trust these predictions?

Opening the black box with SHAP and LIME.



What Drives the Model? (Validating Physics with SHAP)



Key finding:

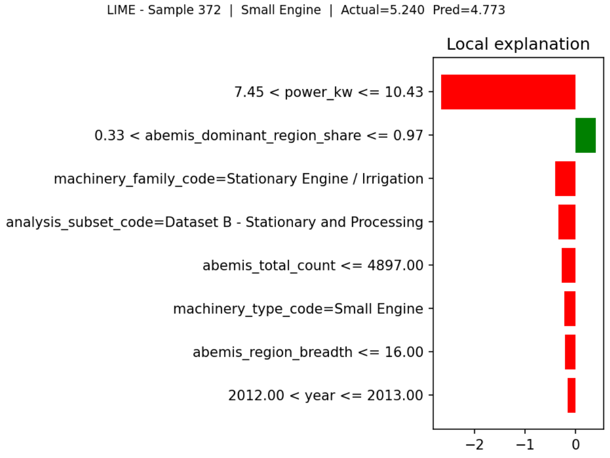
`power_kw` accounts for ∼**66.5%** of total prediction importance.

`year` (equipment vintage) is second at ∼**12.3%**, reflecting improved combustion efficiency in newer equipment.

ABEMIS context features contribute a combined ∼**9%**, adding real but secondary value.

The model didn't memorize data — it *learned physics*.

Explaining Individual Predictions (LIME)



LIME breaks down *why* the model predicted **4.77 L/h** for one specific small engine — granular transparency that builds trust for BAFE engineers.

From Lab to National Scale

172,265

machines successfully scored

National inventory:	246,741
Coverage:	**70 %**
Missing power data:	62,808
Unmappable type:	11,668

Outputs per region:

- Per-record fuel estimate (L/h)
- Per-region aggregate demand
- Per-region $\times$ machinery-type breakdown

The unscored 30% reflects *data gaps* — not model failure. Closing those gaps is part of the roadmap.

What’s next?

Closing the data gaps. Validating in the field.

Next Iterations

PDF

Expand Corpus

Acquire **3,065-PDF** collaborator set; Tesseract OCR already deployed — 267 scanned reports processed, 376-record corpus achieved.

CAL

RAED Calendar

Integrate cropping-season data to add **temporal load timing** as a predictive feature.

$\sim$

Field Telemetry

Validate predictions against **in-field sensor data** from a pilot deployment.

Thank you

Ronald Melvin R. Rosas

Marcus Rafael Tiongson

AI 201 — University of the Philippines, Diliman